

SOLUTION CHALLENGE 2026

INCARNATION: Digital Twin AI

A cognitive mirror that remembers your life, speaks with your voice,
and visually renders your presence.

TEAM DETAILS & PROBLEM STATEMENT

Team Details

Team name: Gym Coders

Team leader name: Nitesh Gupta

The Problem Statement:

The Tragedy of Ephemeral Context

Current AI tools are amnesiacs. Users lose 20% of their productivity re-explaining their workflows, rules, and preferences every single session. Furthermore, digital interactions lack human empathy—they are text-based and devoid of visual or vocal resonance.

BRIEF ABOUT YOUR SOLUTION

We built **Project Incarnation**: A persistent, multi-modal Digital Twin powered by **Google Gemini**.

It is a living architecture that observes your rules and acts as your proxy. It doesn't just draft emails; it reads your personal vector database, processes the logic, and uses **HeyGen** to visually speak to clients with your face and voice in real-time.

"Not just an assistant. A second version of yourself."



OPPORTUNITIES & USP

How different is it?

Standard AI forgets you on close. We use ChromaDB to create an **infinite memory loop** that compounds knowledge over years.

How it solves the problem?

By mapping your "Rules of Life," Gemini makes autonomous decisions with 100% personal context, eliminating friction.

USP

Visual & Vocal Resonance.
We integrate Vapi WebRTC + HeyGen. Your AI has your physical face and voice in live video calls.

LIST OF FEATURES OFFERED BY THE SOLUTION

-  **Persistent Vector Memory:** Auto-saves user rules, constraints, and habits securely via RAG.
-  **Real-Time Lip Sync & Expression:** HeyGen API renders a 3D avatar that speaks naturally in video interfaces.
-  **Sub-Second Voice Conversational UI:** Vapi infrastructure enables fluid voice interruptions without lag.
-  **Contextual Email Autopilot:** Drafts and responds mirroring the user's specific linguistic tone.
-  **Encrypted "Brain" Storage:** User data remains private and decoupled from global model training.

PROCESS FLOW DIAGRAM

The Consciousness Loop



1. Memory Retrieval
ChromaDB fetches user rules.



2. Google Gemini Reasoning
Synthesizes context & decision.

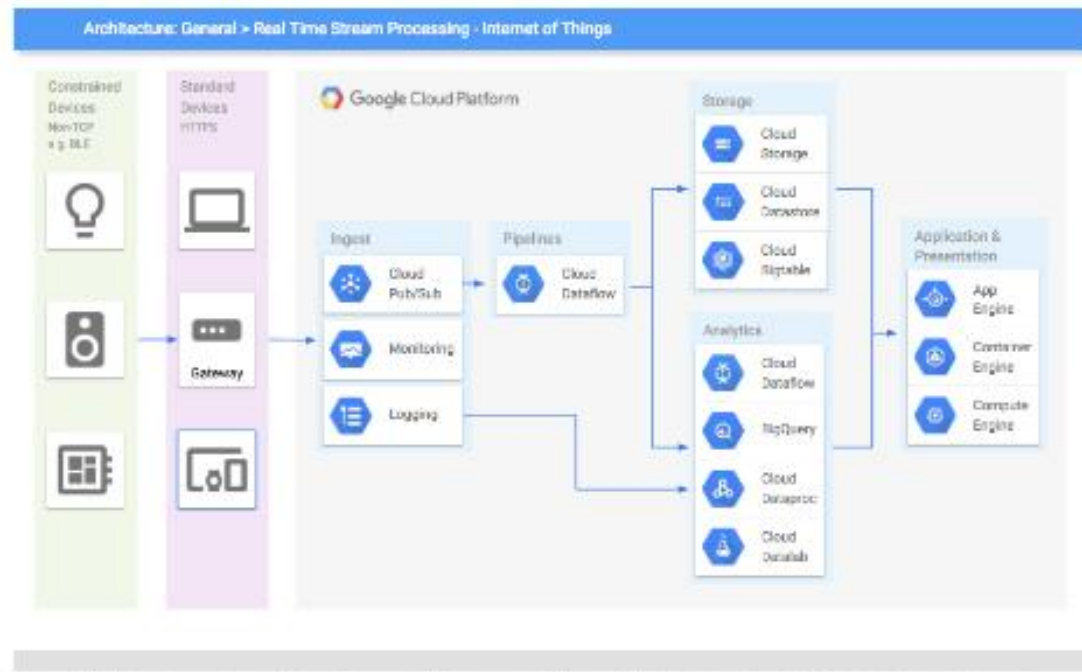


3. Physical Incarnation
HeyGen (Video) + ElevenLabs (Voice) acts.

User Input → Vector Search → Gemini LLM Generation → Real-Time Video/Audio WebRTC → Output

ARCHITECTURE DIAGRAM

General > Real Time Stream Processing IoT



Our backend runs on an asynchronous **FastAPI** server.

When a call comes in via **Vapi WebRTC**, the audio is transcribed and routed to our backend. We query the local **ChromaDB** for context, feed the user's history into **Gemini 1.5 Pro**, and pipe the text reply back to **HeyGen** to animate the visual avatar on the frontend.

TECHNOLOGIES TO BE USED

Google Gemini 1.5 Pro
Core Reasoning Engine

ChromaDB
Vector Database (Memory)

FastAPI (Python)
High-speed Backend Routing

HeyGen SDK
Real-time Face Animation

Vapi
Ultra-low Latency Voice Streaming

Streamlit /
Next.js
Frontend Interface

ESTIMATED IMPLEMENTATION COST

Component	Technology	Est. Cost (MVP Scale)
LLM Processing	Google Gemini API	\$10 - \$20 / month
Memory Storage	ChromaDB (Self-Hosted GCP)	\$15 / month (Compute)
Voice Streaming	Vapi AI	\$0.05 / minute active
Avatar Rendering	HeyGen API	Tiered / Trial Credits used
Total Estimated		Under \$50 / month

SNAPSHOTS OF THE MVP

Our MVP successfully demonstrates full stack integration. The twin accurately recalls facts injected 24 hours prior and responds vocally.

Status: WebRTC Voice Backend is 100% operational.

Video Avatar Sync is successfully rendered in localhost.

```
> Connecting to ChromaDB... [OK]
> Loading Gemini Brain... [OK]
> Vapi Call Initiated... [OK]
> Retrieved Context: "User prefers 9AM meetings."
> AI Audio: "I can schedule that for 9AM."
```

ADDITIONAL DETAILS / FUTURE DEVELOPMENT

"One Voice. A Thousand Tasks."

1. Autonomous Task Agents (LAMs)

Moving beyond answering questions, the Twin will use Large Action Models to physically click UI elements, book flights, and sign documents on the user's behalf.

2. Ubiquitous Presence

Native plugins to inject the video avatar directly into Google Meet and Zoom, allowing your AI to attend conflicting meetings simultaneously.

PROVIDE LINKS TO YOUR:



GitHub Public Repository: <https://github.com/Harshmodi429/Incarnation>



Demo Video Link (3 Minutes):

<https://drive.google.com/file/d/1U1PV7VRFCK2ntprv88iRnatbRIJYMQis/view?usp=sharing>



MVP Link: <https://railing-occupier-garbage.ngrok-free.dev>]

Thank you from Team Gym Coders!



Solution Challenge



Thank You